

MOONTASIR RAFIQUE

◇ mrafi013@ucr.edu ◇ [LinkedIn](#) ◇ [Google Scholar](#) ◇

PROFESSIONAL SUMMARY

PhD researcher in ECE specializing in **wireless communications, physical-layer security, and AI/ML for next-generation wireless systems**. Proficient in **MATLAB, Python, and PyTorch** for link-level simulation, ML model development, and performance evaluation. Developed domain-adaptive RIS beam selection and conditional generative models for LEO satellite anomaly detection. Research spans AI/ML-enabled wireless systems, probabilistic shaping, RF channel modeling, 3GPP-informed simulation, and secure communications.

WORK EXPERIENCE

UC Riverside: Hua's Lab | Graduate Student Researcher

06/2024–Present

- Conducting **DoD-funded** research on wireless physical-layer security; developed Python/MATLAB pipelines for large-scale signal simulation, interference modeling, and performance metric extraction (channel capacity, SER, outage probability).
- Modeled distributed adversaries (**eavesdroppers and jammers**, including PPP-based eavesdropper geometry) and developed Python-based Monte Carlo simulation workflows to evaluate secrecy capacity and secure-link performance across varying SNR, channel conditions, and modulation schemes.
- Quarterly technical presentations at **DoD** project reviews; one conference paper published (MILCOM 2025).

UC Riverside | Teaching Assistant

09/2024–12/2024, 06/2025–07/2025, 03/2026–06/2026

- **EE111 (Digital and Analog Signals and Systems)**: Designed and taught MATLAB labs covering FFT-based spectral analysis, FIR/IIR filter design, Fourier/Laplace/Z-transforms, modulation/demodulation, and communication system simulation; mentored students in MATLAB programming and debugging.
- **EE115 (Introduction to Communication Systems)**: Led laboratory sessions for 30+ undergraduates covering Simulink-based communication system modeling, digital filter design, and LTspice/PSpice circuit simulation.

SKILLS

Wireless & Digital Communications: Digital signal processing (DSP), digital communication systems, OFDM, NOMA, finite constellation, physical-layer security, constellation shaping (CS), RF channel modeling, link-level simulation and performance analysis, LEO satellite communications, reconfigurable intelligent surfaces (RIS), and practical familiarity with selected 3GPP specifications (TR 38.821, TS 38.133, TS 38.331).

Machine Learning & AI: Domain-adversarial neural networks (DANN), variational autoencoders (VAE/cVAE), normalizing flows, anomaly detection, semi-supervised domain adaptation, transfer learning, convolutional neural networks (CNN), multilayer perceptrons (MLP); PyTorch.

Programming & Automation: Python (NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, PyTorch), MATLAB, NVIDIA Sionna Library (LDPC/channel coding).

Analysis Tools: Spectrum Analyzer, Signal Generator, Oscilloscope, GNU Radio.

Simulation & Design: CST Microwave Studio, Cadence Virtuoso (SpectreRF), LTspice, PSpice, Simulink, Atoll, AutoCAD, QuaDRiGa (3GPP geometry-based stochastic channel simulation).

Data Analysis: Excel (*Pivot Tables, VLOOKUP, Conditional Formatting*), Tableau.

EDUCATION

- **Ph.D. in Electrical and Computer Engineering (ECE)** 09/23 – Continued
-University of California at Riverside; Advisor: Professor Yingbo Hua
- **M.S. in Electrical and Computer Engineering (ECE)** 09/2023 – 12/2024
-University of California at Riverside; Grade: 3.90/4.00

RELEVANT COURSES

Graduate Level: EE224: Digital Communication, EE226: Wireless Communication, EE236: State and Parameter Estimation Theory, EE221: RF Integrated Circuit Design, EE231: Convex Optimization EE230: Advanced Linear Algebra

Undergraduate Level: Digital Signal Processing, Mobile Communication, Wireless Communication, Electronics I & II, Power Electronics.

SELECTED PROJECTS

Physics-Based LEO Satellite Channel Simulator — MATLAB, Python, PyTorch

- Developed a physics-based LEO satellite channel simulator modeling a two-shell constellation (60 satellites, 8 ground stations) using Keplerian + J2 propagation, tropospheric refraction, and a 3GPP TR 38.821 SC8 Ka-band link budget.
- Generated per-timestep handover telemetry including SINR, Doppler shift, handover latency, handover rate, and recovery time, incorporating measurement assumptions from 3GPP TS 38.133.
- Implemented physically grounded handover, beam-tracking, and jamming anomalies for ML benchmarking, incorporating 3GPP TS 38.331 handover procedures and propagation-based attack models.

Conditional Generative Anomaly Detection for LEO Satellite Networks — Python, PyTorch

- Developed conditional VAE (cVAE) and conditional Normalizing Flow (cFlow) models for unsupervised anomaly detection on real Starlink drive-test telemetry comprising 22,958 labeled samples.
- Incorporated mobility and geometric context into the generative models, improving anomaly detection performance to **AUROC = 0.916** with cVAE and **0.914** with cFlow, compared with **0.751** for an unconditional VAE.
- Achieved strong detection across scheduling failures, beam obstruction, and complete link failures, including **82.1% TPR at 10% FPR** for scheduling anomalies and over **99% TPR** for complete link failures.

Domain-Adaptive Beam Selection for RIS via Domain Adversarial Neural Network (DANN) — Python, PyTorch

- Developed a DANN framework for RIS beam selection under LOS-to-NLOS domain shift, with MLP and CNN backbones trained on 3GPP-compliant geometry-based stochastic channel datasets across static and dynamic UE scenarios.
- Showed that unsupervised domain alignment causes negative transfer in the MLP (23.25% → 20.67%), while semi-supervised adaptation with 30% labeled target data recovers accuracy to **49.08%** on the static scenario and **43.75%** on the dynamic one, gains of **25.83** and **29.00** percentage points over no-adaptation baselines.
- Demonstrated MLP superiority over the 1D CNN across nearly all settings, with SSDA performance gaps of up to **13.41 pp** at 30% annotation, attributing the difference to feature representation suitability. *Submitted to IEEE SCSETech 2026.*

Information Theoretic Security and Physical-Layer Design for STEEP — Python, PyTorch, MATLAB, Sionna, GPU

- Developed finite-constellation models for the STEEP physical-layer security protocol using asymmetric PSK, APSK, QAM, and PCS-QAM across different modulation orders $M = 2-256$, deriving and evaluating key capacity and secrecy rate for different encryption (nonlinear and linear) and decision (hard and soft detection) schemes.
- Validated information-theoretic secrecy gains at the coded level by implementing 5G NR LDPC encoding and belief-propagation decoding via the Sionna library, deriving channel LLR expressions for the legitimate receiver and eavesdropper and simulating coded BER/BLER across code rates and modulation orders to identify where theoretical secrecy margins translate into practical performance at finite blocklength.
- Extended STEEP to stochastic wireless threat models with spatially distributed eavesdroppers modeled as a Poisson Point Process (PPP), evaluating secrecy rate and secrecy-outage probability across eavesdropper densities and adversarial cooperation using GPU-accelerated Monte Carlo simulations, and benchmarking against the Wyner wiretap baseline.

PUBLICATIONS: SCHOLARLY WORK

1. **Domain-Adversarial Adaptation for Robust RIS Beam Selection Under Propagation Shifts.** *Submitted to IEEE SCSETech 2026, Pomona, California, USA.*
2. **Investigation of STEEP Using Phase-Shift-Keying Signaling for Secure Communications.** *Presented, MILCOM 2025, Los Angeles.*
3. **Suitability of FPS and DPS in NOMA for Real-Time and Non-Real-Time Applications.** *Proc. EECSI 2021, Indonesia — Best Paper Award & Best Presenter Award.* [\[IEEE Link\]](#)
4. **Impact of Receiver Antenna Polarization and Resource Scheduler on the Downlink Performance of High-Velocity Users in 5G mmWave Small Cell Technology.** *Proc. ICECIT 2021.* [\[IEEE Link\]](#)
5. **Configuring Antenna System to Enhance Downlink Performance of High-Velocity Users in 5G MU-MIMO Networks.** *Proc. ACMI 2021.* [\[IEEE Link\]](#)
6. **Downlink Performance Enhancement of High-Velocity Users in 5G Networks by Configuring Antenna System.** *BIM 2021.* [\[Springer Link\]](#)
7. **Performance Comparison of DPS in NOMA for Different MIMO Antenna Configurations.** *Proc. ICCCE 2023.* [\[IEEE Link\]](#)

AWARDS

- | | |
|--|------|
| • Dean's Distinguished Fellowship Award UC Riverside | 2023 |
| • Best Paper Award ⁽¹⁾ 8th International Conference on EECSI | 2021 |